

2079

Bachelor of Science in Computer Science and Information Technology

Course Title: **Mathematics II**

Code No: **MTH168** Semester: **2nd Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Define system of linear equations. When a system of equations is consistent? Make echelon form to solve:
$$\begin{cases} -2x_1 - 3x_2 + 4x_3 = 5 \\ x_2 - 2x_3 = 4 \\ x_1 + 3x_2 - x_3 = 2 \end{cases}$$
is consistent.
2. Define linear transformation with an example. Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $v = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and define a transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(x) = Ax$ then (a) find $T(v)$ (b) find $\{x \in \mathbb{R}^2\}$ whose image under T is $\{b\}$.
3. Find AB by block multiplication of the matrices $A = \begin{bmatrix} 2 & -3 & 1 & 0 & -4 \\ 1 & -5 & -2 & 3 & -1 \\ 0 & -4 & -2 & 7 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 6 & 4 \\ 2 & -1 \\ -3 & 7 \\ 1 & 3 \\ 5 & -3 \end{bmatrix}$

Section B

Attempt any Eight questions [8×5=40]

4. Find the least square solution of $Ax = c$, where $A = \begin{bmatrix} 1 & -3 & -3 \\ 1 & 5 & 1 \\ 1 & 7 & 2 \end{bmatrix}$, $c = \begin{bmatrix} 5 \\ -3 \\ 5 \end{bmatrix}$ and compute the associated least square error.
5. Determine the column of the matrix A are linearly independent, where $A = \begin{bmatrix} 3 & -3 & 6 \\ 2 & 4 \\ 0 & 3 & 0 \end{bmatrix}$
6. Let $A = \begin{bmatrix} 1 & 5 \\ -3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -5 \\ 3 & k \end{bmatrix}$. What value (s) of k , if any, will make $AB = BA$?
7. Evaluate the determinant of the matrix $\begin{bmatrix} 1 & -7 & 8 & 9 & -6 \\ 0 & 2 & -5 & 7 & 3 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & -1 & 0 \end{bmatrix}$
8. When two column vectors in $\mathbb{R}^2$ are equal? Given an example. Compute $(u + 3v, -u - 2v)$, where $u = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$, $v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.
9. Prove that the two vectors u and v are perpendicular to each other if and only if the line through u is perpendicular bisector of the line segment from $-u$ to v .

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10. Find the eigenvalue of $A = \begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix}$.
11. Define null space of a matrix A . Let $A = \begin{bmatrix} 1 & -3 & 2 \\ -5 & 9 & -1 \end{bmatrix}$, and $v = \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix}$, then show that v belongs to the null space matrix A .
12. Find the equation $y = a_0 + a_1 x$ of the least squares line that best fits the data points $(2, 1)$, $(5, 2)$, $(7, 3)$, $(8, 3)$